Problem 6-8

The purchased-equipment cost for a plant which produces pentaerythritol (solid-fluid processing plant) is $300,000. The plant is to be an addition to an existing formaldehyde plant. The major part of the building cost will be for indoor construction. The contractor's fee will be 7% of the direct plant cost. All other costs are close to the average values found for typical chemical plants. On the basis of this information, estimate the total direct plant cost, the fixed capital investment, and the total capital investment.

Solution:

This solution is based on Example 6-1 and uses the "colorful worksheet." Solve the colorful worksheet first, using cost component factors for solid-fluid processing plants. Adjust the factors to account for indoor construction in an existing plant. Adjust the contractor's fee percentage until the contractors fee is 7% of the direct plant costs.

	A	B	C	D	E	F	G	H	I	J	K
1	ESTIMATION OF CAPITAL INVESTMENT BY PERCENTAGE OF DELIVERED EQUIPMENT METHOD										
2	(See Table 6-9)										
3	The fractions in the cells below are approximations applicable to typical chemical processing										
4	plants. These values may differ depending on many factors such as location, process type, etc.										
5	Required user input			Default		Subtotal		Result			
6	Required, from a linked sheet or entered manually						Notes & comments				
7	Project Identifier: Illustration 101			Fraction of delivered equipment			User: copy	Calculated			
8				Solid-	Solid-fluid	Fluid	from values	values,			
9				processing	processing	processing	at left or	million $			
10				plant	plant	plant	insert				
11				Direct Costs							
12		Purchased equipment, E'						0.300			
13		Delivery, fraction of E'		0.10	0.10	0.10	0.10	0.030			
14		Subtotal: delivered equipment						0.330			
15		Purchased equipment installation		0.45	0.39	0.47	0.39	0.129			
16		Instrumentation&Controls(installed)		0.18	0.26	0.36	0.26	0.086			
17		Piping (installed)		0.16	0.31	0.68	0.31	0.102			
18		Electrical systems (installed)		0.10	0.10	0.11	0.10	0.033			
19		Buildings (including services)		0.25	0.29	0.18	0.32	0.106	Constraint: Increase building		
20		Yard improvements		0.15	0.12	0.10	0.12	0.040	cost for indoor construction		
21		Service facilities (installed)		0.40	0.55	0.70	0.55	0.162			
22		Total direct costs		1.69	2.02	2.60	2.05	1.007	←ANS		
23											
24				Indirect Costs							
25		Engineering and supervision		0.33	0.32	0.33	0.32	0.106			
26		Construction expenses		0.39	0.34	0.41	0.34	0.112	Constraint:		
27		Legal expenses		0.04	0.04	0.04	0.04	0.013	Contractor's fee is 7% of FCI		
28		Contractor's fee		0.17	0.19	0.22	0.2135	0.070	7.0000 =100*H28/H22		
29		Contingency		0.35	0.37	0.44	0.37	0.122	Manipulate G28 until J21 is 7		
30		Total indirect costs		1.28	1.26	1.44	1.28	0.424			
31											
32		Fixed capital investment (FCI)					ANS→	1.430	Sent to 'Evaluation' and		
33									'Year-0 $', there adjusted		
34		Working capital (WC)		0.70	0.75	0.89	0.75	0.248	as described below		
35											
36		Total capital investment (TCI)						1.678	←ANS		

Problem 6-9

Estimate by the turnover ratio method the fixed capital investment required in 2000 for a proposed sulfuric acid plant (battery-limit) which has an annual capacity of 1.3×10^8 kg/yr of 100 percent sulfuric acid (contact catalytic process), using the data form Table 6-11, when the selling price for the sulfuric acid is \$86 per metric ton. The plant will operate 325 days/year. Repeat the calculation, using the cost capacity exponent method with data from Table 6-11.

Solution:

Method 1 - Turnover Ratio

$$\text{Turnover Ratio} = \text{TOR} = \frac{\text{Gross Annual Sales}}{\text{Fixed Capital Investment}} = \frac{G_j}{\text{FCI}} = \text{constant}$$

From Table 6-11 contains data for the sulfuric acid plant.

$$\text{TOR}_1 = \text{TOR}_2$$

$$\left(\frac{G_j}{\text{FCI}}\right)_1 = \left(\frac{G_j}{\text{FCI}}\right)_2$$

$$\frac{9\times10^7\ \frac{\text{kg}}{\text{y}}\cdot\frac{\$86}{1000\text{kg}}}{\$4,000,000} = \frac{1.3\times10^8\ \frac{\text{kg}}{\text{y}}\cdot\frac{\$86}{1000\text{kg}}}{\text{FCI}}$$

$$\text{FCI} = \underline{\underline{\$5,778,000}}_{\text{ANS}}$$

Method 2 – Cost capacity exponent method (scaling):

$$\text{Cost of Plant A} = \text{Cost of Plant B}\cdot\left(\frac{\text{Capacity of Plant A}}{\text{Capacity of Plant B}}\right)^n$$

$$\text{Cost of Plant A} = \$4,000,000\cdot\left(\frac{130,000,000\ \text{kg/yr}}{90,000,000\ \text{kg/yr}}\right)^{0.65} = \underline{\underline{\$5,080,000}}_{\text{ANS}}$$

Problem 6-10

The total capital investment for a chemical plant is $1 million, and the working capital is $100,000. If the plant can produce 8000 kg of final product per day during a 365-day year, what selling price in dollars per kilogram of product would be necessary to give a turnover ratio of 1.0?

Solution:

$$\text{TCI} = \$1,000,000 \text{ and } \text{WC} = \$100,000$$

$$\therefore \text{FCI} = \text{TCI} - \text{WC} = \$1,000,000 - \$100,000 = \$900,000$$

$$\text{Turnover Ratio} = \text{TOR} = \frac{\text{Gross Annual Sales}}{\text{Fixed Capital Investment}} = \frac{G_j}{\text{FCI}} = 1.0$$

$$\frac{\frac{8000\text{ kg}}{\text{day}} \cdot \frac{365\text{ days}}{\text{yr}} \cdot \left(x\,\frac{\$}{\text{kg}}\right)}{\$900,000} = 1.0$$

Solving for x gives x=$0.308/kg

ANS

Lesson 16 – In-Class Problem – Breakeven Analysis

1. Objectives
2. Problem Statement.
3. Suggested Approach and Definitions.
4. Submission Requirements.

Objectives

1. Learn how to prepare a breakeven chart.
2. Learn how to measure the breakeven point.
3. Calculate the return on investment given economic data.
4. Apply economic data to determine an optimal production rate.

Problem Statement

The table below contains production data for a chemical processing plant with a total capital investment of $17,940,000. The plant operates for 8,000 hrs/y. From the data, determine:

a) the production rate at the breakeven point,
b) the maximum net annual profit (N_p) and corresponding (optimum) production rate,
c) the maximum return on investment (ROI) and corresponding production rate,
d) the product market value in $/kg at the optimum production rate.
e) the variable production costs in $/kg at the optimum production rate.

Production Rate, kg/s	Gross earnings, $/s	Variable costs, $/s	Fixed costs, $/s
0.0	0.000	0.591	0.380
5.0	0.603	0.842	0.380
10.0	1.206	1.089	0.380
15.0	1.809	1.390	0.380
17.5	2.110	1.621	0.380
20.0	2.411	1.969	0.380

Suggested Approach and Definitions

- Much of the information needed may be found on pages 231-232 of the PTW text. Refer to these pages for additional definitions.
- Return on investment (ROI) is defined on page 323.
- Be creative.

Submission Requirements

None, but confirm your work with your instructor.

Solutions

"Visual" or quick solutions from inspection of graph:

a) ~14.5 kg/s
b) ~$0.109/s at 17.5 kg/s
c) ~17.5% at 17.5 kg/s.
d) ~$0.12/kg at 17.5 kg/s
e) ~$0.093/kg at 17.5 kg/s

More precise solutions from interpolation of regression trendline:

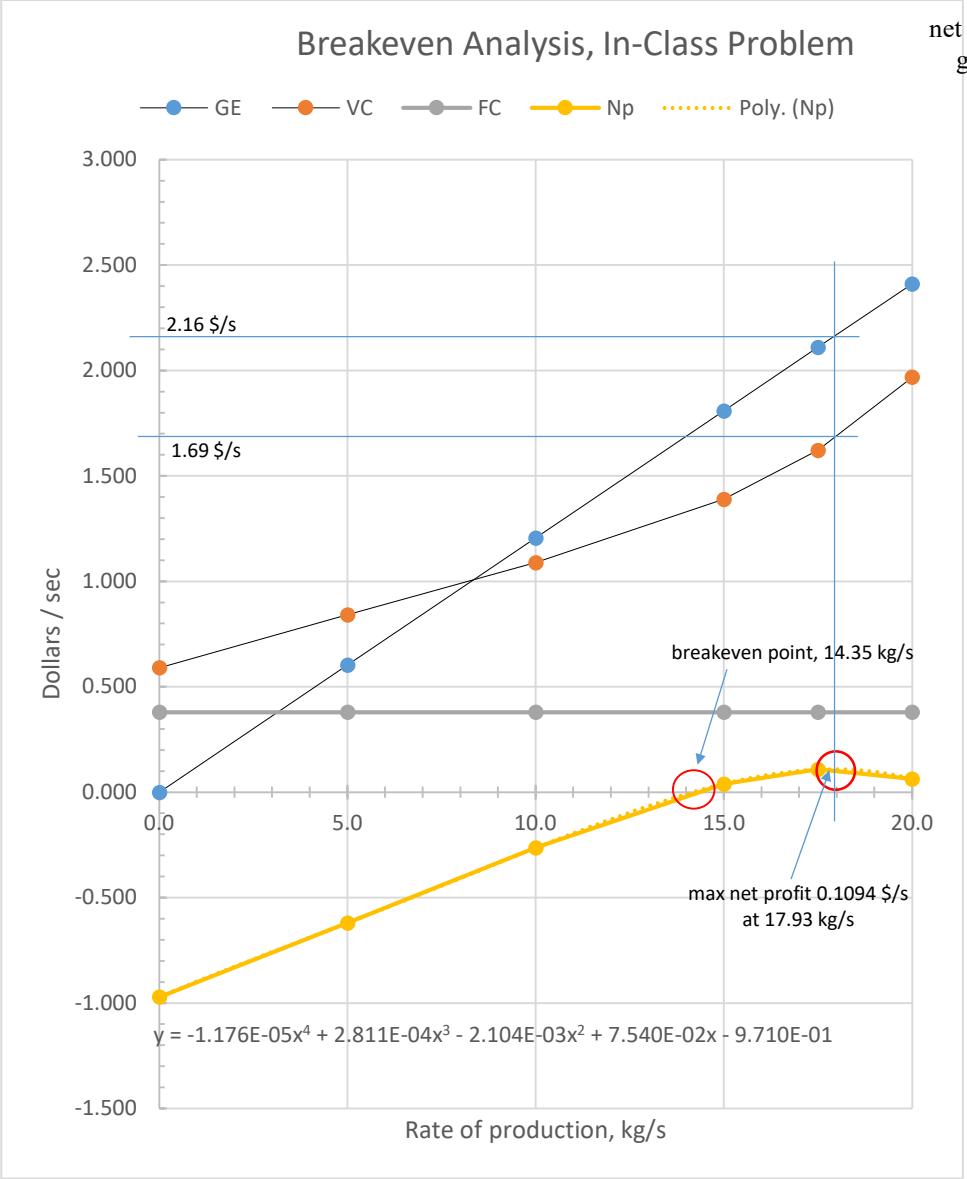
f) 14.35 kg/s
g) $0.1094/s at 17.93 kg/s
h) 17.6% at 17.93 kg/s.
i) $0.121/kg at 17.93 kg/s
j) $0.094/kg at 17.93 kg/s

Work is shown on the following page.

APPROVED SOLUTION

PR, kg/s	GE, $/s	VC, $/s	FC, $/s	Np, $/s	ROI
0.0	0.000	0.591	0.380	-0.971	-1.559
5.0	0.603	0.842	0.380	-0.619	-0.994
10.0	1.206	1.089	0.380	-0.263	-0.423
15.0	1.809	1.390	0.380	0.039	0.062
17.5	2.110	1.621	0.380	0.109	0.175
20.0	2.411	1.969	0.380	0.062	0.100

PR = production rate
GE = gross earnings
VC = variable costs
FC = fixed costs
Np = net profits
ROI = return on investment

ROI =
net profit ÷ TCI

net profit =
gross profit – costs